Exploring Risk Factors and Predicting UPDRS Score Based on Parkinson's Speech Signals

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***Abstract*—The unified Parkinson's disease rating scale (UPDRS) is the most widely employed scale for tracking Parkinson's disease (PD) symptom progression. However, conventional way to achieve UPDRS, mainly based on the physical examinations of clinic patients performed by the trained medical staffs, involves the disadvantages of inconvenience and high medical expense. Hence, in this study, we try to explore some risk factors and accurately predict the UPDRS for PD, using the speech signals of PD patients published on UCI machine-learning archive. More specifically, inspired by the idea of ensemble learning, we firstly construct a framework of ensemble feature selection (EFS) to select a suitable subset of features among numerous speech signals. Subsequently, a personalized predictive model, trained by adopting information from similar patients, is developed to be customized for an individual PD patient. Finally, we employ the personalized predictive model to predict UPDRS score combined with various classical regression algorithms. Compared to conventional models, our study has a potential to capture more relevant risk factors and produces more accurate UPDRS score for individual patient. Experimental results on real-world dataset from UCI machine-learning archive show that our personalized predictive model gets a promising performance.**

Keywords-UPDRS; speech signals; framework of ensemble feature selection; personalized predictive model

I. INTRODUCTION

Parkinson's disease (PD) is a long-term degenerative disorder of the central nervous system that mainly affects the motor system. In the early stages, the symptoms including tremor, rigidity, slowness of movement, and difficulty with walking, talking, thinking or completing other simple tasks. Dementia becomes common in the advanced stages of the disease. Depression and anxiety are also common occurring in more than a third of people with PD, other symptoms include sensory, sleep, and emotional problems [1].

In 2017, PD affected more than 10 million people worldwide [2], making it the second most common neurologic condition after Alzheimers disease. From the male-to-female ratio perspective, males are one and a half times more likely to have PD than females [2]. And from the age, the PD typically occurs in people over the age of 60, of which about one percent are affected [3]. Additionally, the costs of PD to society and personal are high. In terms of the United States alone, the combined direct and indirect cost of PD, including treatment, social security payments and lost income from inability to work, is estimated to be nearly $25 billion per year. As for individual person with PD, medication and therapeutic surgery can cost up to $102,500 dollars [2]. Nevertheless, the cause of PD is not clear, but believed to involve both genetic and environmental factors. PD traditionally has been considered a non-genetic disorder, however, around 15% of individuals with PD have a first-degree relative who has the disease [4]. At least 5% of people are now known to have forms of the disease that occur because of a mutation of one of several specific genes [5]. Recently a number of environmental factors have been associated with an increased risk of PD, including pesticide exposure, head injuries, and living in the country or farming [6]. What's worse, There is no cure for PD. Initial treatment is typically with the antiparkinson medication L-DOPA (levodopa), but as the disease progresses and neurons continue to be lost, these medications become less effective [7].

PD symptom severity is typically quantified using clinical metrics. One of the most commonly used PD metrics is the Unified Parkinson's Disease Rating Scale (UPDRS) [8]. For untreated patients, UPDRS spans the range 0 to 176, with 0 representing healthy state and 176 representing total disabilities, and consists of six sections [9]: 1) evaluation of mentation, behavior and mood. 2) self-evaluation of the activities of daily life including speech, swallowing, handwriting, dressing, falling, walking, and cutting food. 3) clinician-scored monitored motor evaluation. The motor UPDRS ranges from 0 to 108, with 0 denoting symptom free and 108 denoting severe motor impairment. 4) complications of therapy. 5) Hoehn and Yahr staging of severity of Parkinson's disease. 6) Schwab and England ADL scale. Doctors often use UPDRS to track PD symptom progression in clinic. However, physical frequent visits of many persons with Parkinson (PWP) to the clinic for monitoring involves inconvenience and expensive medical expenses for patients [10]. Fortunately, the current advancement in the telecommunication industry allows nowadays to considering the possibility of telemonitoring [11], that offers the remote monitoring of such patients for real time diagnosis. It has been suggested that speech is affected very early [12] and continues to deteriorate as the disease progresses. Therefore the speech signal is a natural candidate for measuring and quantifying the progress of the disease. In fact, there are many traditional and novel methods to extract the PD's speech signals and these algorithms has extracted numerous speech signal features including Jitter, Shimmer, PRDE, PPE, SNR, HNR, MFCC and other speech features [13]. In our research, the purpose of the study is twofold: 1) to explore a suitable subset of distinctive risk factors among numerous speech signals. 2) to build a personalized predictive model to predict UPDRS score. The model is customized for an individual patient and trained using information from similar patients [14].

The remainder of this paper is organized as follows. Section II summarizes some works related to PD prediction and some methods of feature selection. Methodology is given in Section III and experiment design is presented in Section IV. The results and model evaluation are discussed in Section V. Finally, we present the conclusions in Section VI.

II. RELATED WORK

It has been proven that approximately 90% of the patients[10] show vocal disorders that affect their speech. And the difficulty in reliable PD diagnosis has inspired researchers to develop data mining methods relying on algorithms aiming to assess progress of PD. Recently, researchers have focused on the prediction of clinical rating scales evaluating severity of PD and its progression [15] [16]. Our study aims to follow the trend and proposes a personalized predictive model that use a suitable subset of distinctive risk factors to achieve

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a promising performance. Many researches have discussed the data mining approaches on PD.

For many data mining tasks, feature selection is a fundamental tool, which selects a small set of important features to improve the performance and efficiency. Nevertheless, feature selection is a difficult task due mainly to a large search space, where the total number of possible solutions is 2^n for a dataset with n features [17]. A variety of search techniques have been applied to feature selection, such as complete search, greedy search, heuristic search, and random search [18]. Sakar and Kursun [19] applied a mutual information-based feature selection algorithm with the permutation test for assessing the relevance and the statistical significance of the relations between the features and the PD-score, and used a leave-one-subject-out (LOSO) cross-validation scheme to avoid bias. They compared the performance of the classifiers using accuracy, sensitivity, specificity and the MCC.

Another study that apply feature selection methods (including LASSO, Relief, LLBS, mRMR) greatly increased the accuracy of classification. The most significant and discriminative features of speech signals were obtained and explained with a medical background [20].

For assessing the PD progression better. Many researches have discussed the potential approaches of data mining. Nicolas Genain et al. [21] used machine learning methods to predict continuous measures of Parkinson's Disease Severity from voice recordings of patients. In their study, two datasets are analyzed, and smoothed by a moving average was applied in random forest. Finally they used an Bagged decision algorithms to improve RMSE.

Tsanas et al. [22] used speech data to predict tendency for progression of disease in which they extracted voice features using signal processing algorithms from 6000 samples of 42 PWP and identified the useful features. Selected subsets of features were mapped to UPDRS using regression and classification techniques.

Shreya Jain and Dr. Sujala Shetty [23] aimed to build a predictive model that accurately predicts the UPDRS of patients using the data collected through noninvasive speech tests. In their study, on one hand, they summaries the PD – symptoms, risk factors, telemonitoring, rating scale. On the other hand, they present a two-step model that predictions to be done with more accuracy than the researches have done before this.

Zoltan Galaz et al. [24] investigated 50 PD patients (UPDRS III ranged from 6 to 92) that comprised 42 speech tasks. The task was divided into subsets: words, sentences, reading text, monologue and diadochokinetic. Then they used guided regularized random forest algorithm to select features with maximum clinical information and performed random forests regression to estimate PD severity.

Michal Vadovsk and Jn Parali [25] are focused on patients classification based on their speech signals. They employed C4.5, C5.0, Random Forest and CART algorithms to create the decision trees. In addition, they cut-off values of individual features were applied in order to classify the patients. The cut-off points obtained based on individual features did not reach very high success rates, which is considered relatively low. Accordingly, the author thinks that the patients' classification using only one feature is not highly accurate and, therefore, numerous features should be considered.

III. METHODOLOGY

In the following, we develop the personalized predictive model to predict UPDRS score. The model is customized for an individuals patient and trained using information from similar patients. Compared to global models trained on all patients, they have the potential to produce more accurate risk score and capture more relevant risk factors for individual patients. To make the model simpler and improve data

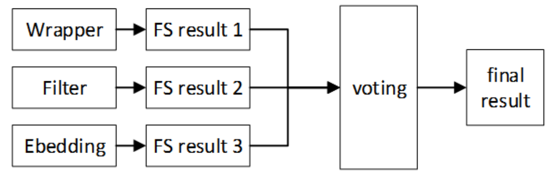

Fig. 1: A framework of ensemble feature selection

mining performance, we learn from the idea of ensemble learning to construct a framework of ensemble feature selection to select a suitable subset of features among numerous speech signals. In order to better show our personalized predictive model, given the training set $X = \{(\vec{x_1}, y_1), (\vec{x_2}, y_2), \cdots, (\vec{x}_m, y_m)\}, \vec{x}_i \in \mathbb{R}^d, y \in \mathbb{R}$, $\vec{x}_i = (\vec{x}_i^1, \vec{x}_i^2, \cdots, \vec{x}_i^d)^T$, $(\vec{x}_i, y_i)$ represents a sample, and there are m samples. The vector $\vec{x_i}$ contains d features.

A. Feature selection

In general, the feature selection (FS) methods can be divided into three categories: Filter, Wrapper and Embedding.

- filter type methods select variables regardless of the model. They are based only on general features like the correlation with the variable to predict [26].
- wrapper type methods evaluate subsets of variables which allows, unlike filter approaches, to detect the possible interactions between variables [27]. In general, since wrapper methods is optimized directly for a given learner, the wrapper methods is better than the filter methods from the final learner performance, but the computational overhead of the wrapper methods is usually much larger than the filter methods due to the wrapper methods need to train the learner several times during the feature selection.
- embedding type methods have been recently proposed that try to combine the advantages of both previous methods. A learning algorithm takes advantage of its own variable selection process and performs feature selection and classification simultaneously. LASSO [28] is a commonly used embedded feature selection method, and the general formulation can be written as Eq.1.

$$\min_{x} \sum_{i=1}^{m} (y_i - \vec{w}^T \vec{x}_i)^2 + \lambda \|\vec{w}\|_1 \tag{1}$$

Where the regularization parameter $\lambda > 0$. λ can alleviate the over-fitting problem, and easily get a sparse solution. And $\vec{w}$ represents the weight vector of the features.

In the process of feature selection, we learn from the idea of ensemble learning. First of all, three feature selection methods(filter, wrapper and embedding) are taken as the basic "estimator", then the best feature subset is selected by voting method. Fig.1 shows our framework of ensemble feature.

B. Personalized predictive model

The conventional approaches to predict UPDRS score focus on a single "global" predictive model using all the available training data. According to Overby C. L. et al. [29], they argued that a static "global" model captures information that is important for the entire training patient population but may miss less popular information that is important for individual patients. Additionally, although some researches have achieved a better performance in predicting UPDRS score, they have used a very complex fusion models or some time-consuming data processing methods. And it will become very bad as the exposition of the data set. Taking into account all aspects, our study builds a personalized predictive model for each patient. In our personalized predictive model, similar PD patients were divided into a cluster according to their own features, and finally trained by using similar patient to predict a new PD's UPDRS score. The model has the potential to generate more accurate UPDRS score and to identify more relevant and informative patient-specific risk factors. In order to ensure the simplicity of the personalized predictive model and the predictive performance of UPDRS score, we finally chose the k-means algorithm to bring together similar patients. In addition, according to the [22] study, we know that the CART algorithm works better. As a result, we combine the k-means algorithm with the CART algorithm.

- *Cluster analysis* In cluster analysis, the k-means algorithm is used to partition the input data set into k partitions (clusters). In fact, the parameter k is known to be hard to choose when not given by external constraints. Considering the sample set $D = \{\vec{x}_1, \vec{x}_2, \cdots, \vec{x}_m\}$, k-means clustering aims to partition the m samples into $k(\leqslant m)$ sets $C = \{C_1, C_2, \cdots, C_k\}$ so as to minimize the within-cluster sum of squares (WCSS). Formally, the objective is to find Eq.2.

$$E = \sum_{1}^{k} \sum_{\vec{x} \in C_i} \|\vec{x} - \vec{\mu}_i\|_2^2 \tag{2}$$

  Where $\vec{\mu}_i = \frac{1}{|C_i|} \sum_{\vec{x} \in C_i} \vec{x}$ is the mean vector of C_i. Intuitively, the Eq.2, to a certain extent, depicts the tightness of the sample in the cluster and the mean vector. The smaller the E, the higher the similarity in the cluster. Nevertheless, minimizing Eq,2 is not easy to find its optimal solution, which is a NP problem. Therefore, the k-means algorithm uses a greedy strategy to approximate Eq.2. Obviously, there is no additional information to determine parameter k in PD's dataset. Thus, we introduce a method commonly used in engineering project in our study, which called elbow method, to estimate of the relatively reasonable number of clusters. Since the k-means model eventually expects minimize the distance of all the data points to which the cluster belongs, and tends to be stable, we can observe that this value with the trend of k to find the best number of clusters. Under the ideal conditions, this line continue to fall and tend to be gentle in the process, there will be a larger slope of the inflection point, the inflection point corresponds to the beginning of the k value.
- *CART* CART is short for Classification and Regression Tree, which is a well-known decision tree learning algorithm that can handle classification and regression tasks. CART uses Gini index to select the partition attribute, and Gini index can be written as Eq.3.

$$Gini(D) = \sum_{k=1}^{|y|} \sum_{k' \neq k} p_k p_{k'} \tag{3}$$

  Where D represents sample set, and $p_k(k = 1, 2, \cdots, |y|)$ or $p'_k(k' = 1, 2, \cdots, |y|)$ represents the proportion of the k-th(or k'-th) class of samples. Intuitively, $Gini(D)$ reflects the probability that two samples are randomly selected from the dataset D, and their classes are not consistent. Therefore, the smaller the $Gini(D)$, the higher the purity of the data set. Likewise, assuming that the discrete attribute a has V possible value $\{a^1, a^2, \cdots, a^V\}$, if the sample D is divided using attribute a, then V branch nodes will be generated, where the v-th branch node contains all samples in D that have a value of a^V on attribute a, that is, D^V. the Gini index of attribute a is defined as Eq.4.

$$Gini\_index(D, a) = \sum_{v=1}^{V} \frac{|D^v|}{|D|} Gini(D^v) \tag{4}$$

  Thus, in the candidate attribute set A, we choose which attribute to minimize the Gini index as the optimal partition attribute, that is, $a_* = \arg\min_{a \in A} Gini\_index(D, a)$.

C. Overview of Our Approach

In comparison with the conventional approaches to predict UPDRS score, our approach fully exploits the features of a single, using a similar patient with fewer features to achieve a better predictive performance. Fig.2 summarizes the conceptual methodology of our proposed personalized predictive model, consisting of four basic steps: speech signals data pre-processing, ensemble feature selection learning, exploring risk factors and predicting UPDRS score.

IV. Experiment

A. Parkinson's Dataset

The dataset was created by Athanasios Tsanas (tsanasthanasis '@' gmail.com) and Max Little (littlem '@' physics.ox.ac.uk) of the University of Oxford, in collaboration with 10 medical centers in the US and Intel Corporation who developed the telemonitoring device to record the speech signals and has been made available online at UCI machine-learning archive [30]. This dataset is composed of a range of biomedical voice measurements from 42(28 males, 14 females) people with early-stage Parkinson's disease recruited to a six-month trial of a telemonitoring device for remote symptom progression monitoring. The recordings were automatically captured in the patient's homes. There are around 200 recordings per patient, which makes a total of 5875 recordings, each represented with a 20 dimensional feature vector along with two attributes(motor_UPDRS score and total_UPDRS score) to predict. The features in the dataset are diverse: some of them are traditional measures that are based on linear signal processing techniques such as Jitter, Shimmer, NHR and HNR; Some are called non-standard measures that are based on nonlinear dynamical systems theory such as recurrence period density entropy (RPDE), detrended fluctuation analysis (DFA), and pitch period entropy (PPE). The labels and short explanations for the measurements along with some basic statistics of the dataset are given in Table I.

From Table I, we can clearly draw the following informations:

1) There are some unreasonable records of feature test_time. (Considering that the feature test_time is not the patient's own characteristics we will delete it).
2) The motor_UPDRS range of PD patients is between 5.04 to 39.51 and total_UPDRS is between 7.00 to 55.00.

Next, in order to better understand and analyzing multivariate data, we apply the parallel coordinate plot to visualize high-dimensional data in Fig. 3. From the Fig.3, we find that the age of male patients were more concentrated (between 50-80), however the distribution of female patients is relatively wide. There are a part of PD patients in the low age (less than 40 years old) and high age (more than 80 years old). Moreover, there are a large proportion of male patients higher than women in UPDRS score and feature of NHR in men and women in a very different.

In the last stage of data exploration, we use the Pearson correlation analysis method to find that there are still relatively large correlations in the feature subset obtained by Athanasios Tsanas et al[22]. Fig.4 shows the Pearson correlation analysis results, which are the deeper the color, the greater the association between attributes. Obviously,

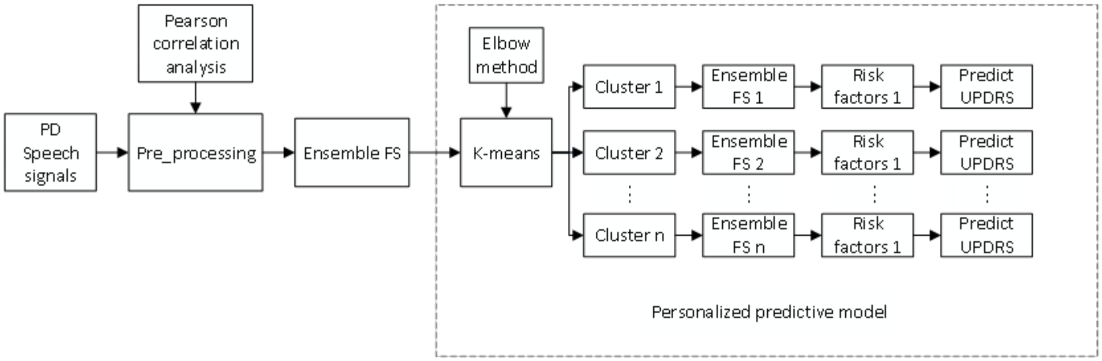

Fig. 2: Methodology plan

TABLE I: DESCRIPTION OF THE FEATURES & TARGETS OF THE PD's DATASET

Features & Targets	Description	Min	Max	Mean	Std
Age	Subject age	36.00	85.00	64.80	8.82
Sex	Subject gender '0' - male, '1' - female	NA	NA	NA	NA
test_time	Time since recruitment into the trial.(day)	-4.26	215.49	92.86	53.45
Jitter(%)	Several measures of variation	0.001	0.100	0.006	0.06
Jitter(Abs)	in fundamental frequency	0.000002	0.0004	0.000044	0.000036
Jitter:RAP		0.000330	0.057540	0.002987	0.003124
Jitter:PPQ5		0.000430	0.069560	0.003277	0.003732
Jitter:DDP		0.000980	0.172630	0.008962	0.009371
Shimmer(dB)	Several measures of variation	0.026000	2.107000	0.310960	0.230254
Shimmer(APQ3)	in fundamental amplitude	0.001610	0.162670	0.017156	0.013237
Shimmer(APQ5)		0.001940	0.167020	0.020144	0.016664
Shimmer(APQ11)		0.002490	0.275460	0.027481	0.019986
Shimmer(DDA)		0.004840	0.488020	0.051467	0.039711
NHR	Two measures of ratio of noise	0.000286	0.748260	0.032120	0.059692
HNR	to total components in the voice	1.659000	37.875000	21.679495	4.291096
RPDF	A nonlinear dynamical complexity measure	0.151020	0.966080 0	0.541473	0.100986
DFA	Signal fractal scaling exponent	0.514040	0.865600	0.653240	0.070902
PPE	A nonlinear measure of fundamental frequency variation	0.021983	0.731730	0.219589	0.091498
motor_UPDRS	Clinician's motor UPDRS score, linearly interpolated	5.04	39.51	21.30	8.13
total_UPDRS	Clinician's total UPDRS score, linearly interpolated	7.00	55.00	29.52	10.7

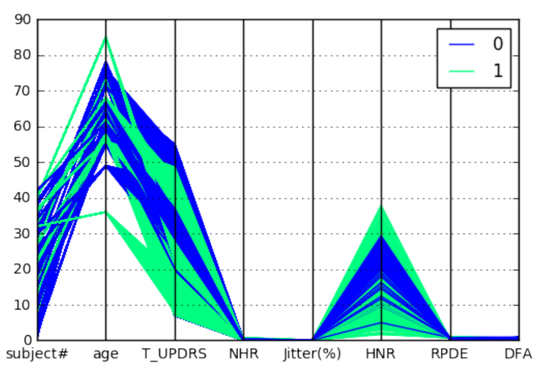

Fig. 3: Parallel coordinates plot of male vs female

Attributes	Jitter(Abs)	Shimmer	NHR	HNR	DFA	PPE
Jitter(Abs)	1	0.649	0.700	-0.706	0.352	0.788
Shimmer	0.649	1	0.795	-0.801	0.133	0.616
NHR	0.700	0.795	1	-0.684	-0.022	0.565
HNR	-0.706	-0.801	-0.684	1	-0.291	-0.759
DFA	0.352	0.133	-0.022	-0.291	1	0.395
PPE	0.788	0.616	0.565	-0.759	0.395	1

Fig. 4: Pearson correlation analysis

we can see that there is a big association between NHR and HNR, Jitter(Abs) and shimmer. Therefore, we believe that there should be a smaller subset of features.

B. Evaluation Metrics

To evaluate the performance of personalized predictive model, we use 10-fold cross validation with the following metrics Eq.5, Eq.6, Eq.7.

$$MAE = \frac{\sum_{i=1}^{m} |y_i - \overline{y}|}{m} \tag{5}$$

$$MSE = \frac{\sum_{i=1}^{m} (y_i - \overline{y})^2}{m} \tag{6}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{m} (y_i - f(\vec{x}_i))^2}{\sum_{i=1}^{m} (y_i - \overline{y})^2} \tag{7}$$

Where $\overline{y}$ represents the prediction result of the personalized predictive model, $f(\vec{x}_i)$ represents the predictive value of the personalized predictive model based on the characteristic vector $\vec{x}_i$. MAE is short for Mean Absolute Error and MSE is short for Mean Squared Error, R^2 not only consider the difference between the regression value and the real value, but also take into account the impact of the real value of the problem itself. In addition, we also use a Sihouette Coefficient to measure the quality of the clustering results. The Sihouette Coefficient also takes into account the degree of cohesion and Separation of the cluster, which is in the range of [-1,1]. The larger the Sihouette Coefficient, the better the clustering effect.

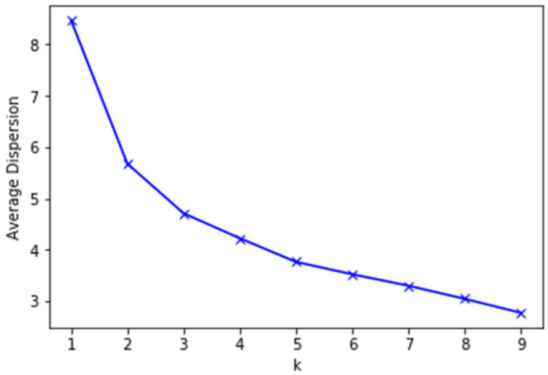

Fig. 5: Selecting k with the elbow method

C. Implementation

After the preliminary analysis we move further and build the personalized predictive model. First, we use the EFS to construct the subset of features needed to predict motor_UPDRS and total_UPDRS. According to Fig.1, we concluded that the risk factors for motor_UPDRS including age, sex, DFA, Jitter(Abs), NHR, HNR and PPE, and the risk factors for total_UPDRS including age, sex, DFA, HNR, RPDE and PPE.

Next, in order to reduce the computational cost of k-means clustering, we use the features chosen by EFS directly. Then we employ elbow method to plot the relationship between the average distance and the number of clusters. As shown in Fig.5, when k=1 or k=2, the average distance of a sample from its cluster falls rapidly, which means that the new number of clusters allows the algorithm to have a larger convergence space, so that the k value can not reflect the real number of clusters. However, when k = 3, the descending velocity of the average distance has a significant slowdown, which means that increase of the k value will no longer favor the convergence of the algorithm, and also implies that k = 3 is relatively optimal number of clusters. Further, we calculate the Sihouette Coefficient and also show that k=3 is a reasonable number of clusters.

Finally, we employ the k-means algorithm to classify PD patients into three categories, and the PD patients in cluster1 including (subject#) 2, 3, 6, 10, 12, 14, 15, 16, 17, 18, 22, 23, 24, 27,33, 34, 20, 36, 37, 38, 39 and 42; Cluster2 including 1, 4, 5, 7, 8, 9, 13, 21, 25, 28, 29, 31, 35, 40 and 41; Cluster3 including 11, 19, 26, 30, 32 and 37. Furthermore, for the task of predictive model, we use Logistic Regression(LR), Least Absolute Shrinkage and Selection Operator(LASSO) and Classification and Regression Tree(CART), which are implemented with python.

V. Results

Now we present the experimental results of ensemble feature selection and our personalized predictive model.

TABLE II shows the performance of our proposed framework of ensemble feature selection in different algorithms. Compared to conventional approaches [22], our proposed method has a slightly improvement in performance. However, the CART algorithm has a huge improvement on the baseline prediction. The results show that the nonlinear algorithm is more suitable for fitting our speech signals.

TABLE III shows the performance of our personalized predictive model in different algorithms. The model is based on our proposed framework of ensemble feature selection, but the prediction performance has been greatly improved in all the clusters. The performance of comparison are shown in Fig.6. Likewise, the CART algorithm presents a great advantage in the personalized predictive model.

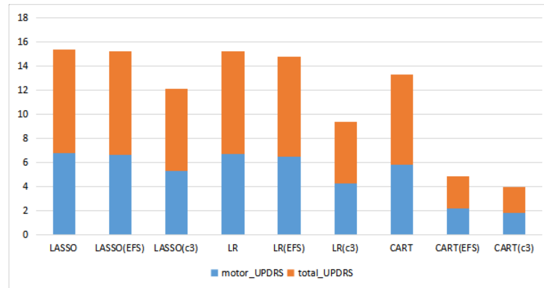

Fig. 6: Performance of 3 regression algorithms using different methods

Hence, we believe our personalized predictive model can produce more accurate UPDRS score and capture more distinctive risk factors among numerous speech signals.

Next, we employ the the framework of feature selection proposed in the methodology to identify the risk factors for each cluster PD patients. As a result, for cluster1, the EFS concluded that the risk factors affecting motor_UPDRS score including age, sex, Shimmer(dB), shimmer:APQ5, HNR and DFA; And the risk factors affecting total_UPDRS score including age, sex, Shimmer(dB), HNR, RPDE and DFA. For cluster2, the EFS concluded that the risk factors affecting motor_UPDRS score and total_UPDRS score including age, sex, HNR, RPDE, DFA and PPE. For cluster3, the EFS concluded that the risk factors affecting motor_UPDRS score and total_UPDRS score including age, sex, Jitter(Abs), Jitter:PPQ5, Jitter:DDP and Shimmer.

TABLE II: PERFORMANCE OF ENSEMBLE FEATURE SELECTION

Method	Target	MAE	MSE	R-squared
LR	motor_UPDRS	6.70	NA	NA
	total_UPDRS	8.50	NA	NA
LASSO	motor_UPDRS	6.80	NA	NA
	total_UPDRS	8.60	NA	NA
CART	motor_UPDRS	5.80	NA	NA
	total_UPDRS	7.50	NA	NA
LR(EFS)	motor_UPDRS	6.45	58.93	0.12
	total_UPDRS	8.31	102.80	0.14
LASSO(EFS)	motor_UPDRS	6.65	61.63	0.08
	total_UPDRS	8.55	107.55	0.10
CART(EFS)	motor_UPDRS	**2.21**	11.40	0.83
	total_UPDRS	**2.66**	16.70	0.86

VI. Conclusions

Our study proposes a personalized predictive model that use a suitable subset of distinctive risk factors to achieve a promising performance. In order to do that, we learn from the idea of ensemble learning to construct an framework of ensemble feature selection to select a suitable subset of features among numerous speech signals. On the one hand, EFS initially found out some risk factors that affect disease; On the other hand, the features chosen by the EFS show a better performance in real-world dataset. Moreover, we move further and build the personalized predictive model.which is customized for an individual PD patient and trained using information from similar patients. Compared to conventional models, our study has a huge improvement, especially in the CART algorithm.

In the future, we plan to collect more disease-related data to test our proposed personalized predictive model. At the same time, we are going to use deep learning methods to study speech signals in

TABLE III: PERFORMANCE OF PERSONALIZED PREDICTIVE MODEL

Method	category	Target	MAE	MSE	R-squared
LR	cluster1	motor_UPDRS	5.23	42.82	0.28
		total_UPDRS	6.78	74.16	0.22
	cluster2	motor_UPDRS	5.27	39.87	0.19
		total_UPDRS	7.45	80.15	0.12
	cluster3	motor_UPDRS	4.25	27.01	0.45
		total_UPDRS	5.12	39.70	0.51
LASSO	cluster1	motor_UPDRS	6.72	60.35	-0.01
		total_UPDRS	8.23	97.01	-0.02
	cluster2	motor_UPDRS	5.75	44.73	0.10
		total_UPDRS	7.81	87.04	0.04
	cluster3	motor_UPDRS	5.27	37.12	0.25
		total_UPDRS	6.88	61.04	0.25
CART	cluster1	motor_UPDRS	1.79	8.65	0.86
		total_UPDRS	2.48	18.00	0.81
	cluster2	motor_UPDRS	2.01	10.36	0.79
		total_UPDRS	2.47	16.36	0.82
	cluster3	motor_UPDRS	**1.80**	9.12	0.81
		total_UPDRS	**2.14**	11.52	0.86

Parkinson patients, so as to avoid a large number of cumbersome artificial extraction of speech signals.

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